

ATV312 Schneider “Modbus RTU”

Siemens S7-1200

Project Overview

Briefly state the goal of establishing *Modbus RTU* communication to control motor speed and direction the goal is avoiding repetitive tasks you can communicate with any vfd and do any function just read vdf manual

Hardware Configuration

List the required components:

Siemens S7-1200 PLC

CM 1241 RS485 communication module

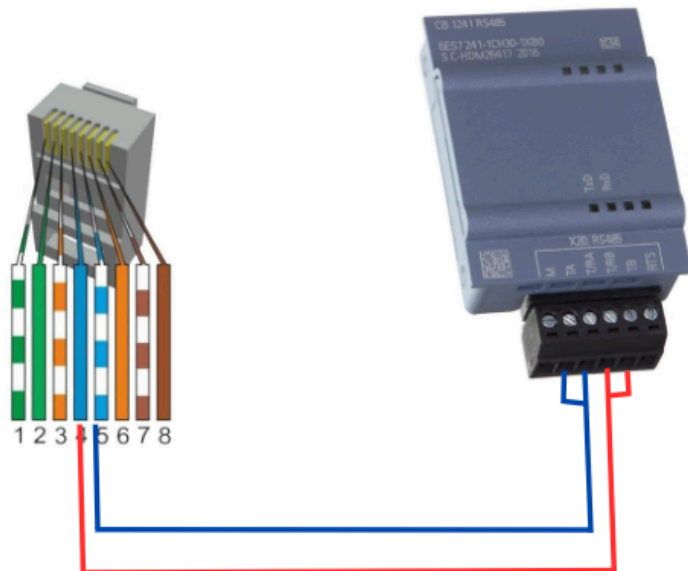
Schneider ATV 312 VFD

Wiring Guide: Detail the *RJ45* connection (Pin 4: Positive, Pin 5: Negative)

ATV 312



S7 1200



VFD Setup Parameters

CtL – Menu (page 58-59-Vfd manual) :

- LAC(Access level) – **L3**
- Fr1(frequency reference1) – **Mdb**
- Fr2(frequency reference2) – **Mdb**
- Cd1(command Run/Direction FWD/REV etc same for cd2) – **Mdb**
- Cd2 – **Mdb**

[COMMAND] (CtL-) menu

Note: There may be an incompatibility between functions (see the incompatibility table, page 21). In this case, the first function configured will prevent the remainder being configured.

Code	Description	Adjustment range	Factory setting
LAC	☐ [ACCESS LEVEL]		[Level 1] (L1)
L1	☐ [Level 1] (L1): Access to standard functions and channel management in order of priority.		
L2	☐ [Level 2] (L2): Access to advanced functions in the [APPLICATION FUNCT.] (FUN-) menu: - +/- speed (motorized jog dial) - Brake control - 2nd current limit switching - Motor switching - Management of limit switches		
L3	☐ [Level 3] (L3): Access to advanced functions and management of mixed control modes		

Fr I	☐ [Ref.1 channel]	[AI1] (AI1)
AI1	☐ [AI1] (AI1) - Analog input AI1	
AI2	☐ [AI2] (AI2) - Analog input AI2	
AI3	☐ [AI3] (AI3) - Analog input AI3	
AIU1	☐ [AI Virtual 1] (AIV1) - In terminal control mode, the jog dial functions as a potentiometer.	
UPdt	If [ACCESS LEVEL] (LAC) = [Level 2] (L2) or [Level 3] (L3), the following additional assignments are possible:	
UPdH	☐ [± SPEED] (UPdt): ± speed reference via LI. See configuration page 78.	
	☐ [±-spd HMI] (UPdH): ± speed reference by turning the jog dial on the ATV312 keypad.	
	To use, display the frequency [Output frequency] (rFr), page 101. The ± speed function via the keypad or the terminal is controlled from the [MONITORING] (SUP-) menu by selecting the [Output frequency] (rFr) parameter.	
LCC	If [ACCESS LEVEL] (LAC) = [Level 3] (L3), the following additional assignments are possible:	
ndb	☐ [HMI] (LCC) reference via the remote display terminal, [HMI Frequency ref.] (LFr) parameter in the [SETTINGS] (SEt-) menu, page 32	
nEt	☐ [Modbus] (Mdb): Reference via Modbus	
	☐ [Com. card] (nEt): Reference via network communication protocol	

Cd I	☐ [Cmd channel 1]	[Terminal] (tEr)
★	Parameter can be accessed if [Profile] (CHCF) = [Separate] (SEP), page 59, and [ACCESS LEVEL] (LAC) = [Level 3] (L3), page 58.	
tEr	☐ [Terminal] (tEr): Control via terminals	
LOC	☐ [Local] (LOC): Control via keypad	
LCC	☐ [Remot. HMI] (LCC): Control via remote display terminal	
ndb	☐ [Modbus] (Mdb): Control via Modbus	
nEt	☐ [Com. card] (nEt): Control via the network	

★ These parameters only appear if the corresponding function has been selected in another menu. When the parameters can also be accessed and set from within the configuration menu for the corresponding function, their description is detailed in these menus, on the pages indicated, to aid programming.

Set menu (page 32-33 in vfd manual)

[SETTINGS] (SEt-) menu

ACC	☐ [Acceleration]	In accordance with Inc , page 63	3 s
	Defined to accelerate from 0 to the nominal frequency [Rated motor freq.] (FrS) in the [MOTOR CONTROL] (drC-) menu.		
dEC	☐ [Deceleration]	In accordance with Inc , page 63	3 s
	Defined to decelerate from the nominal frequency [Rated motor freq.] (FrS) (parameter in the [MOTOR CONTROL] (drC-) menu) to 0. Check that the value for [Deceleration] (dEC) is not too low in relation to the load to be stopped.		
LSP	☐ [Low speed]	0 to HSP	0
	Motor frequency at min. reference		
HSP	☐ [High speed]	LSP to tFr	bFr
	Motor frequency at max. reference: Ensure that this setting is appropriate for the motor and the application.		

COM – Menu (page 98 in vfd manual:

- Add (Address) – 1
- tbr (Baud rate) – 9.6
- tF0 (Com. Type) – 8E1
- tt0 (Timer Error Detection) – 3.0

Code	Description	Adjustment range	Factory setting
Add	☐ [Modbus Address] Modbus address for the drive.	1 to 247	1
tbr 4. 8 9. 6 19. 2	☐ [Modbus baud rate] Modbus transmission speed ☐ [4.8 Kbps] (4.8): 4,800 bits/second ☐ [9.6 Kbps] (9.6): 9,600 bits/second ☐ [19.2 Kbps] (19.2): 19,200 bits/second (Note: This is the only value which supports the use of the remote display terminal.)		19,200 bps
tF0 80 1 8E 1 8n 1 8n 2	☐ [Modbus format] ☐ [8-O-1] (801): 8 data bits, odd parity, 1 stop bit ☐ [8-E-1] (8E1): 8 data bits, even parity, 1 stop bit (Note: This is the only value which supports the use of the remote display terminal.) ☐ [8-N-1] (8n1): 8 data bits, no parity, 1 stop bit ☐ [8-N-2] (8n2): 8 data bits, no parity, 2 stop bits		[8-E-1] (8E1)
tt0	☐ [Modbus time out]	0.1 to 30 s	10 s
AdCd	☐ [CANopen address] CANopen address for the drive.	0 to 127	0
bCd0 10. 0 20. 0 50. 0 125. 0 250. 0 500. 0 1000	☐ [CANopen bit rate] Modbus transmission speed ☐ [10 kbps] (10.0): 10 kbps ☐ [20 kbps] (20.0): 20 kbps ☐ [50 kbps] (50.0): 50 kbps ☐ [125 kbps] (125.0): 125 kbps ☐ [250 kbps] (250.0): 250 kbps ☐ [500 kbps] (500.0): 500 kbps ☐ [1 Mbps] (1000): 1000 kbps		125 bps
ErCd 0 1 2 3 4	☐ [Error code] ☐ No error ☐ Bus off ☐ Life time ☐ CAN overrun ☐ Heartbeat		-

Register Address List

Plc holding register range : For Communication Modbus PLC Siemens S7-1200

PLC Address = "40001 + VFD Address"

MODE	Modbus Function	Data length	Operation and data	Modbus Address
0	01	1 to 2000 1 to 1992 ¹	Read output bits: 1 to (1992 or 2000) bits per request	1 to 9999
0	02	1 to 2000 1 to 1992 ¹	Read input bits: 1 to (1992 or 2000) bits per request	10001 to 19999
0	03	1 to 125 1 to 124 ¹	Read Holding registers: 1 to (124 or 125) words per request	40001 to 49999 or 400001 to 465535
0	04	1 to 125 1 to 124 ¹	Read input words: 1 to (124 or 125) words per request	30001 to 39999
1	05	1	Write one output bit: One bit per request	1 to 9999
1	06	1	Write one holding register: 1 word per request	40001 to 49999 or 400001 to 465535
1	15	2 to 1968 2 to 1960 ¹	Write multiple output bits: 2 to (1960 or 1968) bits per request	1 to 9999
1	16	2 to 123 2 to 122 ¹	Write multiple holding registers: 2 to (122 or 123) words per request	40001 to 49999 or 400001 to 465535
2	15	1 to 1968 2 to 1960 ¹	Write one or more output bits: 1 to (1960 or 1968) bits per request	1 to 9999
2	16	1 to 123 1 to 122 ¹	Write one or more holding registers: 1 to (122 or 123) words per request	40001 to 49999 or 400001 to 465535

Register for Control (vfd communication manual page 19) :

- Set 48502 (dec)= **128** >>> Reset Fault
= **15** >>> Forward Run
= **2063** >>> Reverse Run
= **4111** >>> Stop

Control variables

Modbus address	CANopen address	Code	Read/Write	Name/Description/Possible values
8501	2037 / 2	CMD	R/W	Control word bit 0: "Switch on": active at 1 bit 1: "Disable Voltage": active at 0 bit 2: "Quick Stop": active at 0 bit 3: "Enable Operation": active at 1 bits 4 to 6: Reserved: set to 0 bit 7: Fault state reset: active on rising edge 0 -> 1 bits 8 to 10: Reserved: set to 0 For "Access level" LAC (page 35) = L1 or L2: bit 11 = 0: Forward direction command bit 11 = 1: Reverse direction command bit 12 = 0: No action bit 12 = 1: Stop command depending on the Stt "Stop type" parameter bit 13 = 0: No action bit 13 = 1: Injection stop command bit 14 = 0: No action bit 14 = 1: Fast stop command bit 15: Reserved: set to 0 For "Access level" LAC (page 35) = L3: Factory assignments bit 11 = 0: Forward direction command bit 11 = 1: Reverse direction command bit 12 = 0: No action bit 12 = 1: Stop command depending on the Stt "Stop type" parameter bit 13: No action bit 14: No action bit 15: No action Bits 11 to 15 can be assigned to the following functions: Ramp switching (rPS) Fast stop (FSt) DC injection (DCI) 2 preset speeds (PS2) 4 preset speeds (PS4) 8 preset speeds (PS8) 16 preset speeds (PS16) 2 preset PI references (Pr2) 4 preset PI references (Pr4) Switching for 2nd current limit (LC2) Switching, motor 2 (CHP) External fault (EtF) For example, to use bit 15 to switch the ramp, simply set the "Ramp switching" rPS configuration parameter (page 39) to Cd15.
8502	2037 / 3	LFr	R/W	Frequency reference via the bus (signed value) Unit: - 1 = 0.1 Hz if bit 9 of CMI (page 20) = 0 - 1 ≈ 0.018 Hz (resolution 32767 points = 600 Hz) if bit 9 of CMI = 1
8503	2037 / 4	PISP	R/W	PI regulator reference via the bus Unit: 0.1% Adjustment range: 0 to 1000

Register for Monitoring (vfd communication manual page 98) :

Output Frequency Address = 40001 + 3202 = **43203** (dec)

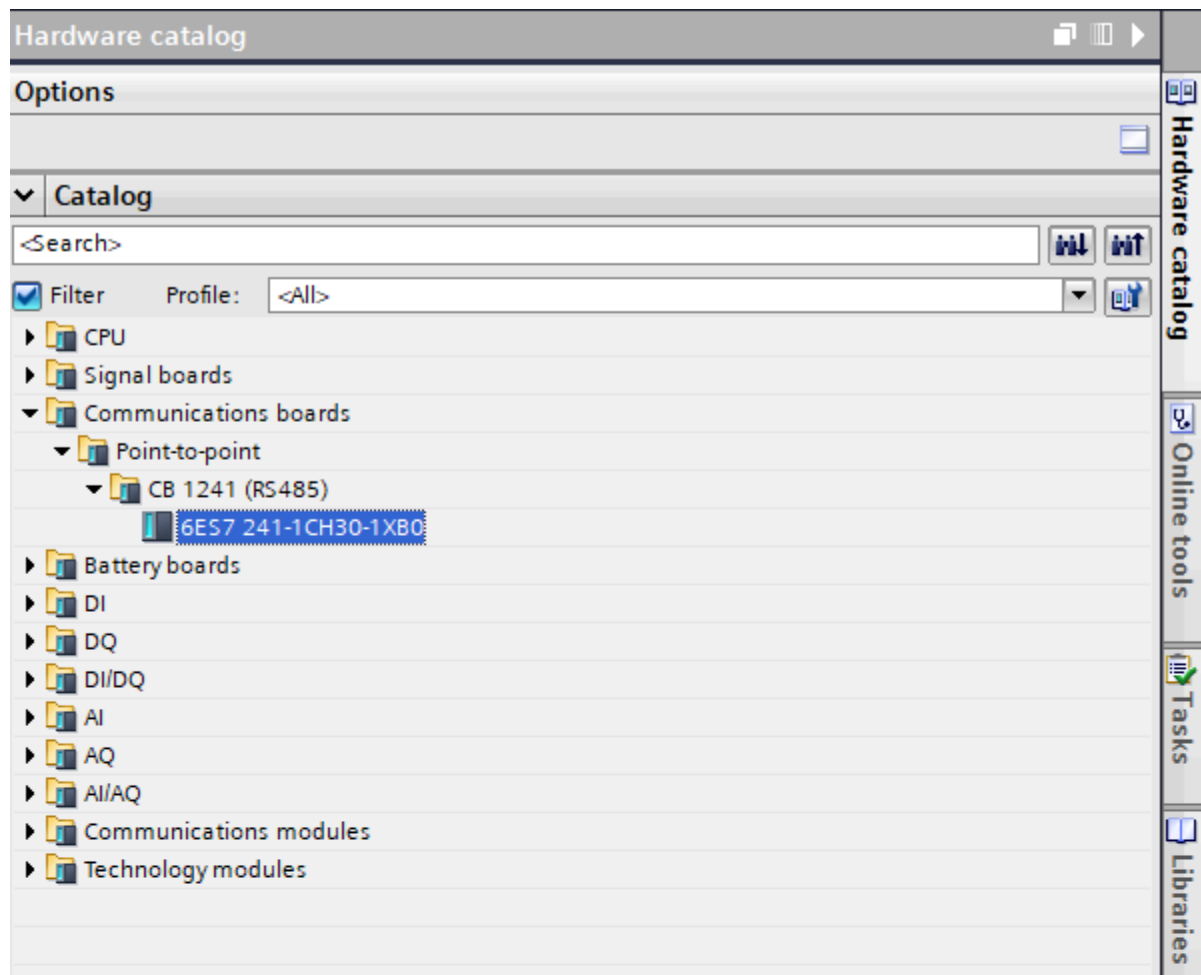
Output Current Address = 40001 + 3204 = **43205** (dec)

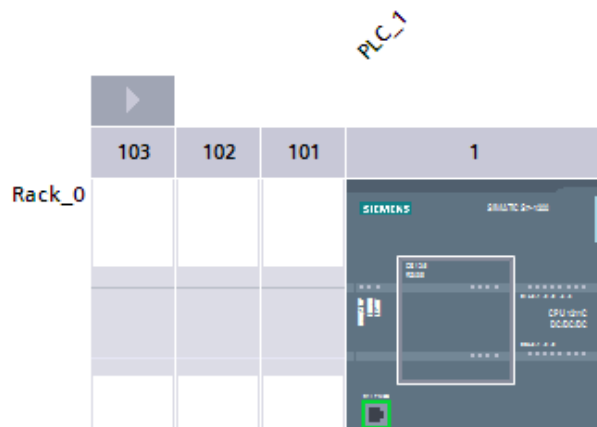
Monitoring variables

Modbus address	CANopen address	Code	Read/Write	Name/Description/Possible values
3201	2002 / 2	ETA	R	Status word bit 0: Ready to switch on bit 1: Switched on bit 2: Operation enabled bit 3 = 0: No detected fault bit 3 = 1: Malfunction, detected fault (FAI) bit 4: Voltage disabled (still equals 0) bit 5: Quick stop bit 6: Switch on disabled bit 7 = 0: No alarm bit 7 = 1: Alarm present bit 8: Reserved bit 9 = 0: Forced local mode in progress (FLO) bit 9 = 1: No forced local mode bit 10 = 0: Reference not reached (transient state) bit 10 = 1: Reference reached (steady state) bit 11 = 0: LFRD reference normal bit 11 = 1: LFRD reference exceeded (< LSP or > HSP) Note: LFRD is expressed in rpm, LSP and HSP in Hz bits 12 and 13: Reserved bit 14 = 0: No stop imposed by STOP key on built-in keypad or on the remote display terminal bit 14 = 1: Stop imposed by STOP key on built-in keypad or on the remote display terminal bit 15 = 0: Forward rotation (output frequency) bit 15 = 1: Reverse rotation (output frequency)
3202	2002 / 3	rFr	R	Output frequency applied to the motor (signed value) Unit: • 1 = 0.1 Hz if bit 9 of CMI (page 20) = 0 • 1 ≈ 0.018 Hz (resolution 32767 points = 600 Hz) if bit 9 of CMI = 1
3203	2002 / 4	FrH	R	Frequency reference before ramp (absolute value) Unit: 0.1 Hz
3204	2002 / 5	LCr	R	Current in the motor Unit: 0.1 A
3205	2002 / 6	Otr	R	Motor torque Unit: 1% 100% = Nominal motor torque, calculated using the configuration parameters.
3211	2002 / C	OPr	R	Motor power Unit: 1% 100% = Nominal motor power, calculated using the configuration parameters.
3207	2002 / 8	ULn	R	Line voltage

Siemens S7-1200 Setup

S7 1200 CB1241 Module Configuration





CB 1241 (RS485) [CB 1241 (RS485)]

General | **IO tags** | **System constants** | **Texts**

► **General**

IO-Link

 Configuration of transmitted ...

► **Configuration of received m...**

IO-Link

Baud rate: 9.6 kbps

Parity: Even parity

Data bits: 8 bits per character

Stop bits: 1

Flow control: None

XON character (HEX): 0

(ASCII): NUL

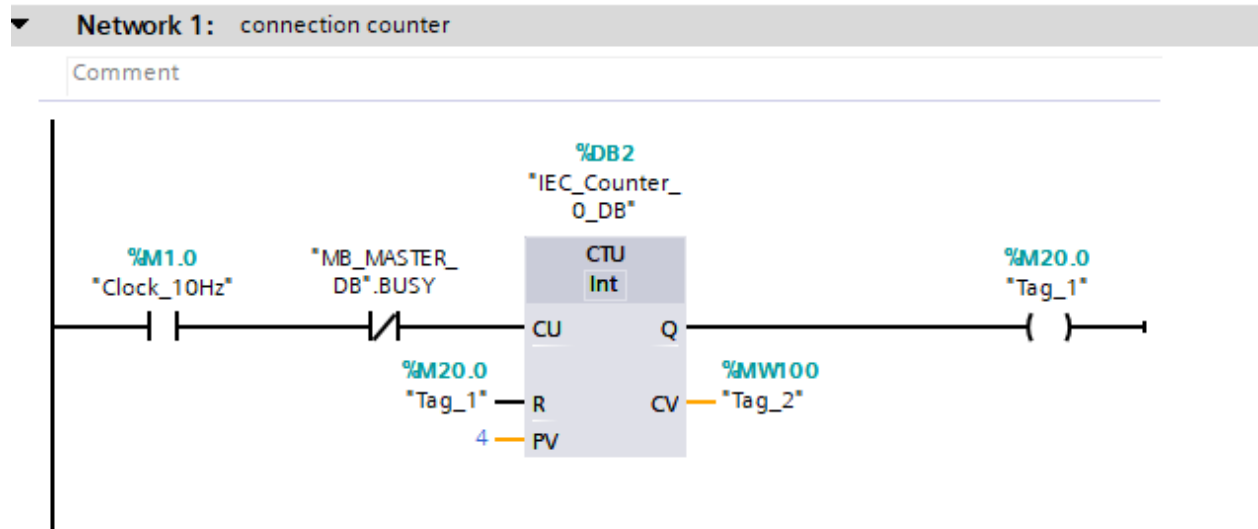
XOFF character (HEX): 0

(ASCII): NUL

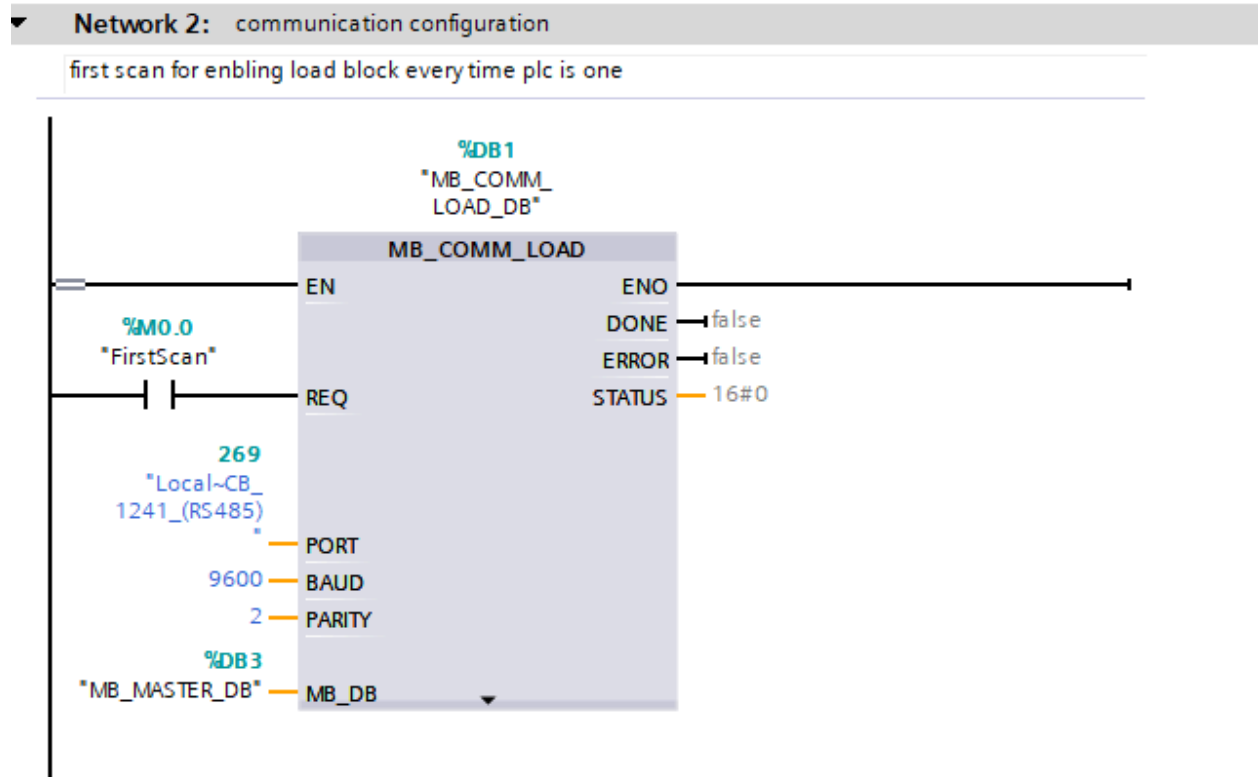
Wait time: 20000 ms

S7-1200 Modbus RTU Programming

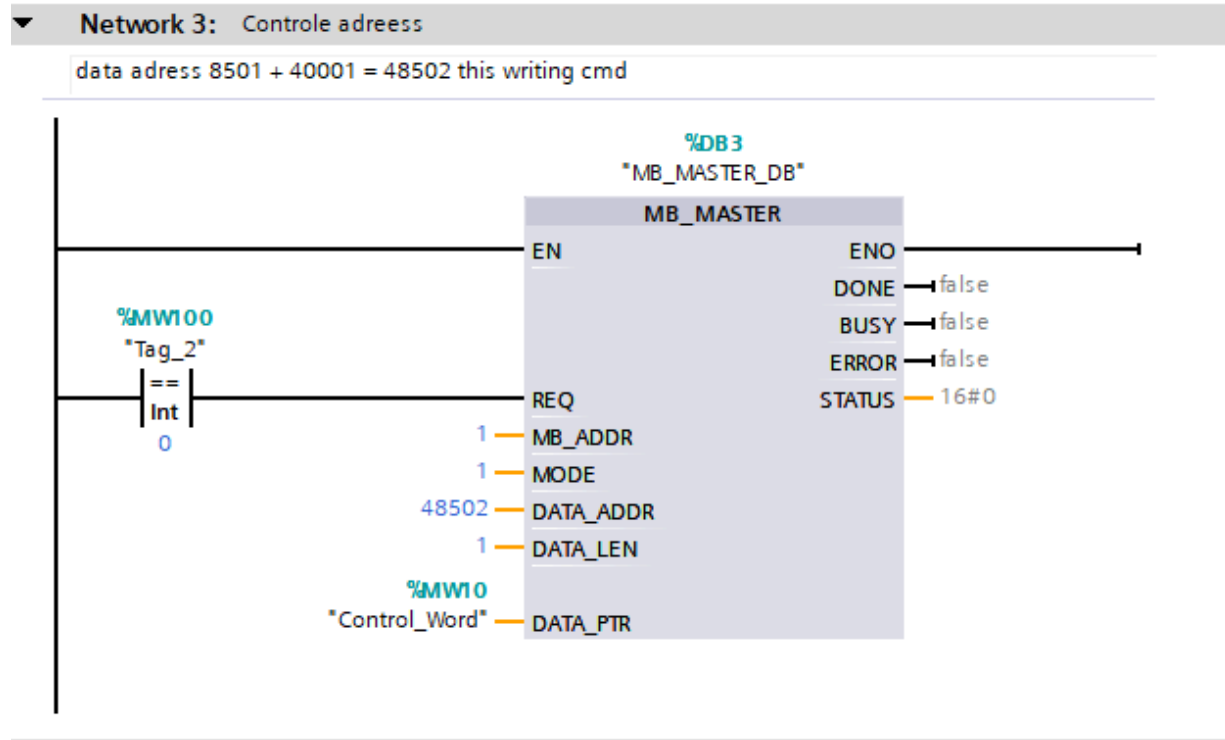
Network 1: Read / Write Distribution



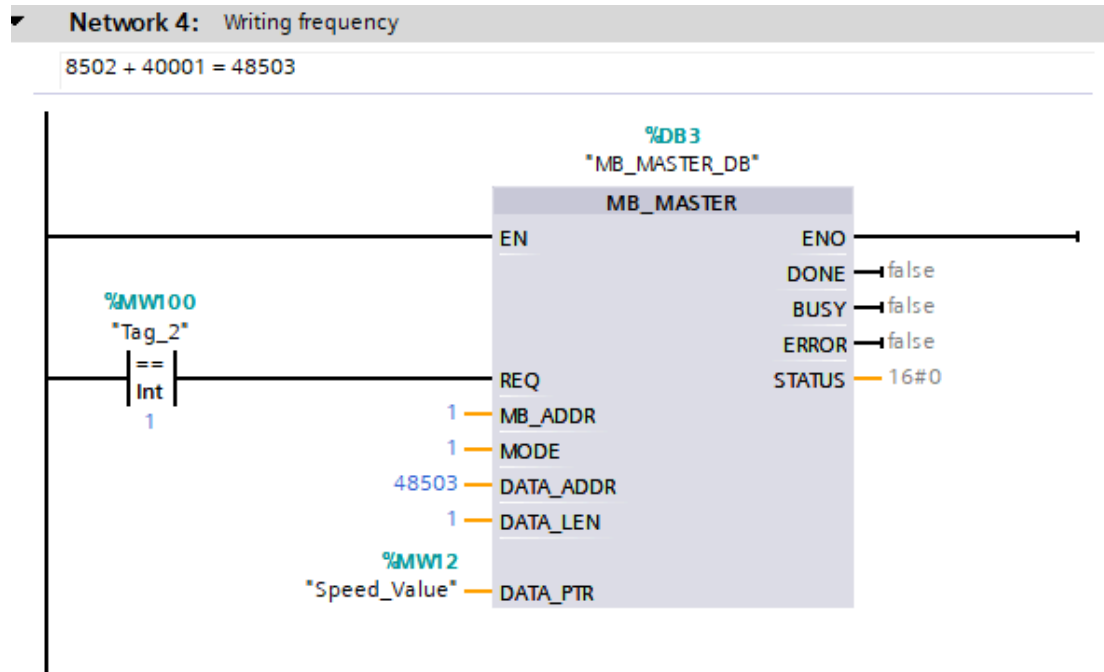
Network 2:Modbus Communication Load



Network 3: Control address

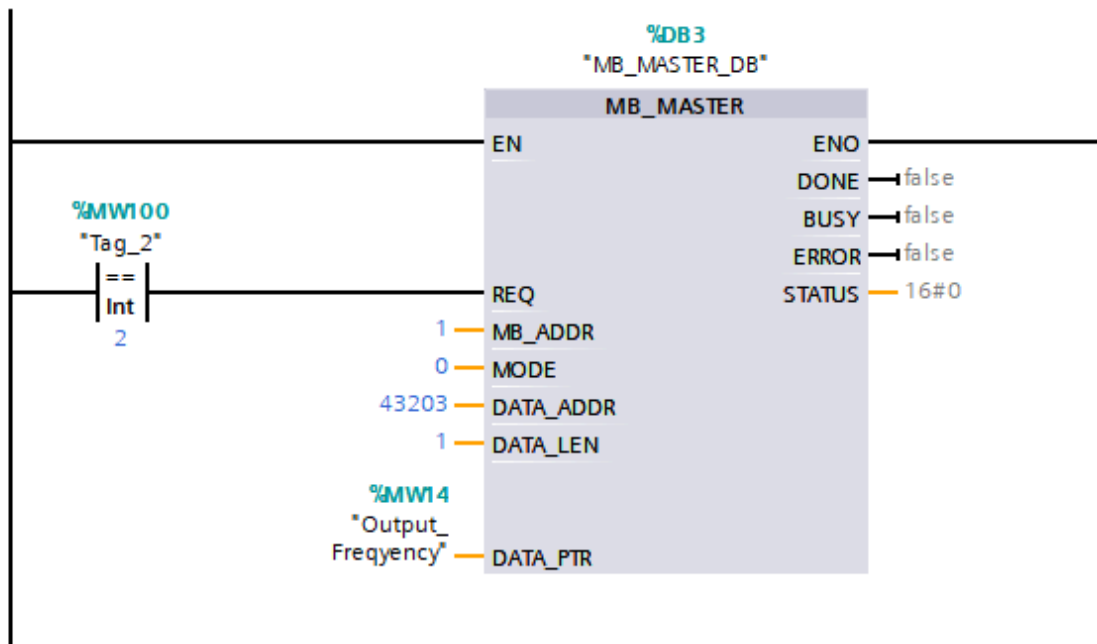


Network 4: Set Frequency Command



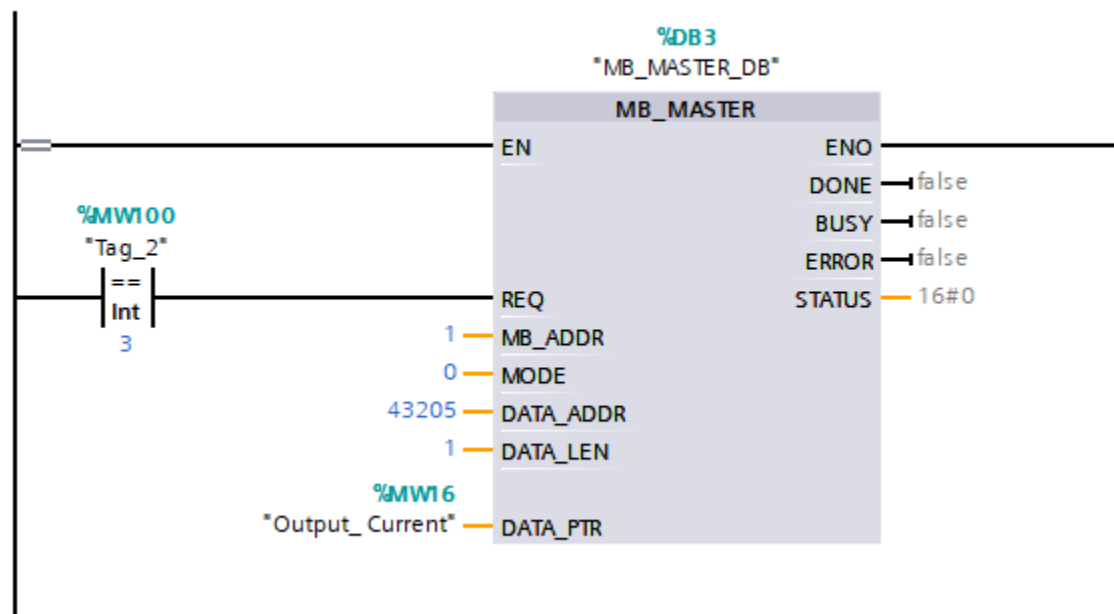
Network 5: reading output frequency

▼ **Network 5:** reading output frequency so mode = 0

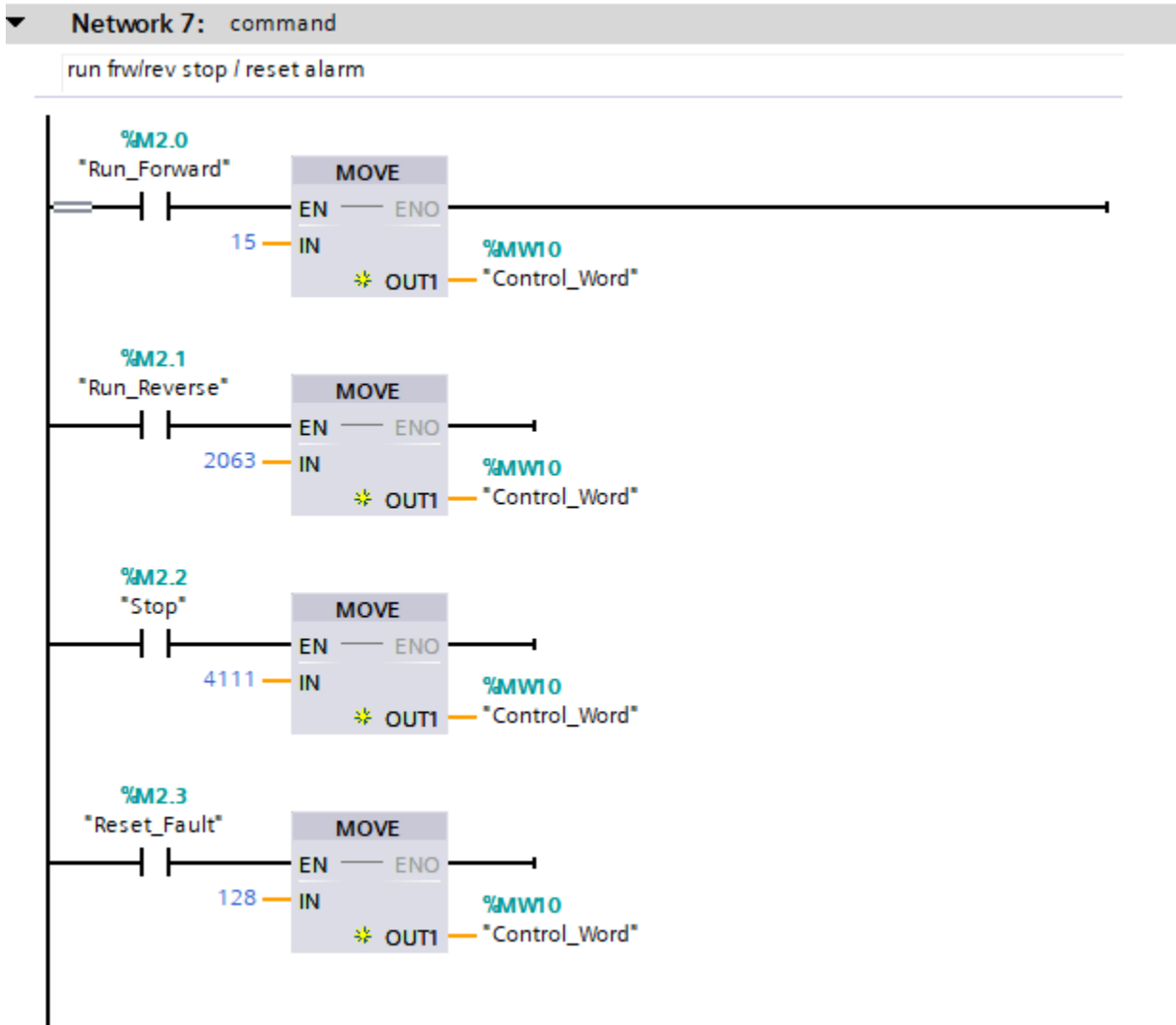
$$4000 + 3202 = 43203$$


Network 5: reading output current

▼ Network 6: reading curent

$$3204 + 40001 = 43205$$


Network 7: Control Motor Command

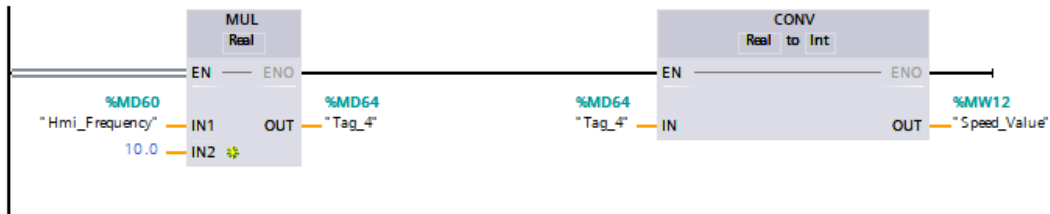


Input data and output data for hmi :

Input : frequency / output : frequency output (actual vfd value), Current output value

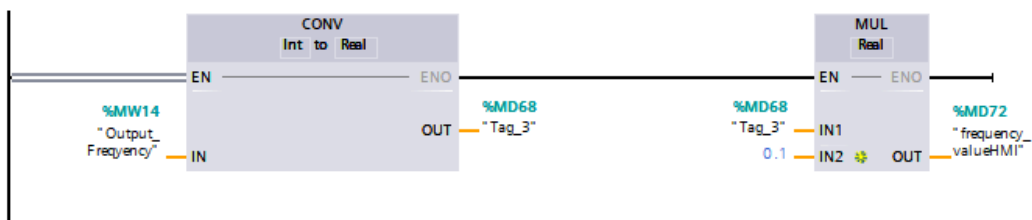
▼ Network 8: frequency hmi input * 10 (from vfd documentation 0.1hz)

Comment



▼ Network 9: display frequency / 10

Comment



▼ Network 10: display current on hmi

Comment

